MAPPA

The first map after the landslide

Consumer-drone photos in. Georeferenced map, change report and blocked roads out.
Offline. On a laptop. No ground control points.

github.com/kavixsri/MAPPA



Responders need a map in 3 hours, not Thursday

Cloud photogrammetry

Needs bandwidth the field does not have.

Ground control points

Nobody is placing GCPs during a disaster.

Satellite tasking

Too slow on day one, and cloud cover can remove it.

Raw video

Footage is not a map. Nobody plans from it.

MAPPA closes the gap: fast, low, offline, and honest about its error.

One offline pipeline, six open-source steps

1

Photos

Interval JPEGs with GPS EXIF

2

Stitch

OpenDroneMap orthophoto + DSM

3

Register

OpenCV SIFT + RANSAC to base imagery

4

Change

Colour diff + DSM−DEM height

5

Roads

GeoPandas / Shapely OSM segments

6

Export

GDAL formats + PDF brief

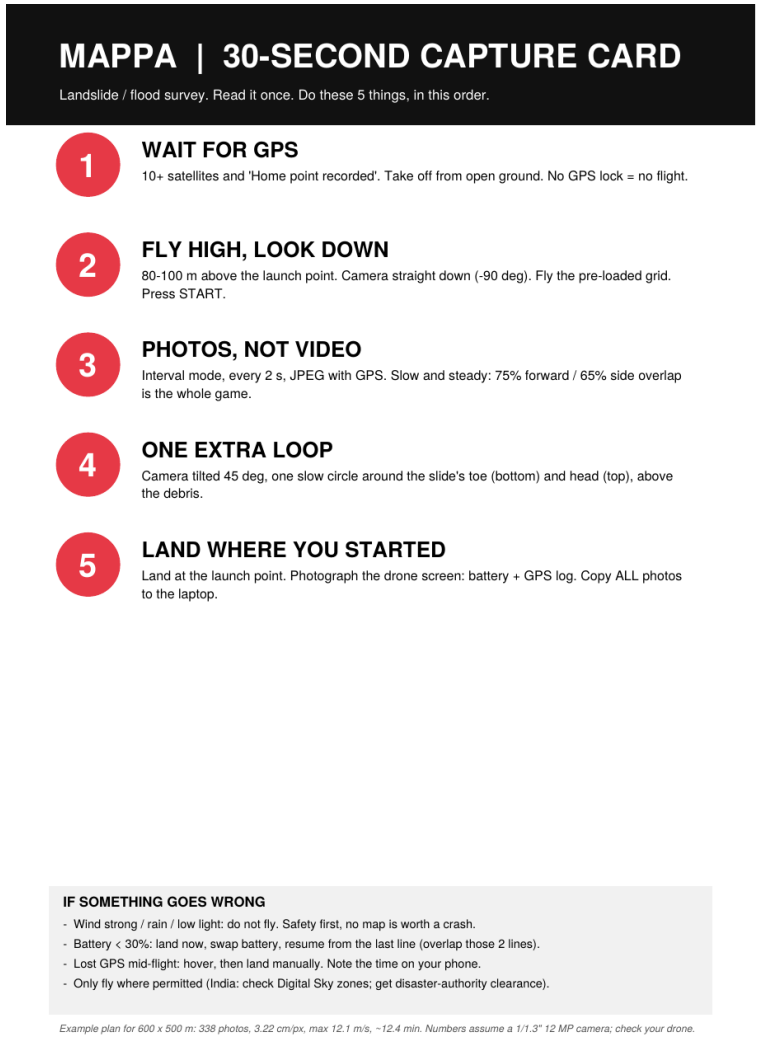
Online once, before any disaster

Sentinel-2 or OpenAerialMap base imagery, OpenStreetMap roads, Copernicus DEM. Stored on the laptop.

Offline in the field

Steps 2 to 6 need no network. The mappa package contains no network imports (grep-verified).

A capture card a stressed pilot can follow



MAPPA | 30-SECOND CAPTURE CARD

Landslide / flood survey. Read it once. Do these 5 things, in this order.

1. **WAIT FOR GPS**
   10+ satellites and 'Home point recorded'. Take off from open ground. No GPS lock = no flight.
2. **FLY HIGH, LOOK DOWN**
   80-100 m above the launch point. Camera straight down (-90 deg). Fly the pre-loaded grid. Press START.
3. **PHOTOS, NOT VIDEO**
   Interval mode, every 2 s, JPEG with GPS. Slow and steady: 75% forward / 65% side overlap is the whole game.
4. **ONE EXTRA LOOP**
   Camera tilted 45 deg, one slow circle around the slide's toe (bottom) and head (top), above the debris.
5. **LAND WHERE YOU STARTED**
   Land at the launch point. Photograph the drone screen: battery + GPS log. Copy ALL photos to the laptop.

IF SOMETHING GOES WRONG

- Wind strong / rain / low light: do not fly. Safety first, no map is worth a crash.
- Battery < 30%: land now, swap battery, resume from the last line (overlap those 2 lines).
- Lost GPS mid-flight: hover, then land manually. Note the time on your phone.
- Only fly where permitted (India: check Digital Sky zones; get disaster-authority clearance).

Example plan for 600 x 500 m: 338 photos, 3.22 cm/px, max 12.1 m/s, ~12.4 min. Numbers assume a 1/1.3" 12 MP camera; check your drone.

1. ## Wait for GPS lock

   10+ satellites, home point recorded.

2. ## 80–100 m, camera down

   Fly the pre-loaded KML grid.

3. ## Photos, not video

   Every 2 s, 75% forward / 65% side overlap.

4. ## One oblique loop

   45° around the slide toe and head.

5. ## Land at launch point

   Photograph the screen. Copy every photo.

Planner: mappa plan gives GSD, speed and image count, and writes the lawnmower KML.
Example 600 × 500 m at 90 m: 338 photos, 3.2 cm/px (assumed 1/1.3" 12 MP camera).

Georeferencing without GCPs, with the error stated

1

Start from camera GPS

ODM places the ortho using EXIF only. Typically metres off.

2

Register to base imagery

SIFT features, RANSAC similarity, applied to the geotransform. Pixels are not resampled.

3

State the bound

acc95 = $\sqrt{(\text{base}^2 + (2 \cdot \text{RMS})^2)}$, printed on every output.

±10 m

Stated absolute accuracy, 95%. Limited by the 10 m base layer. Unverified against ground truth.

5.7 → 0.14 m

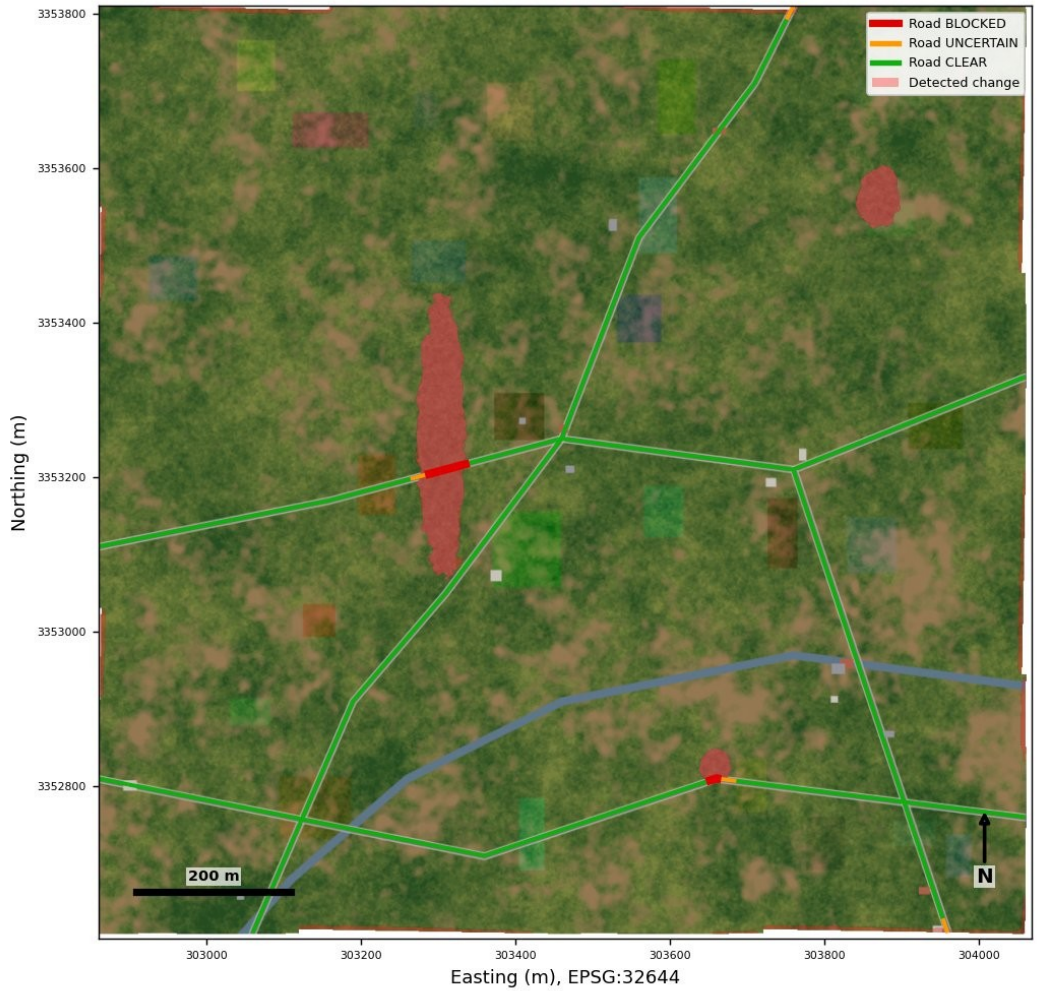
Mean error before → after, on a synthetic scene with a known injected GPS error (25 check points).

0.3–2.7 m

Error after correction across 12 synthetic runs (10 m and 20 m base layers).

Synthetic results show the method works. They are not a field accuracy claim.

What changed, and which roads are blocked



BLOCKED ≥ 40% of the road buffer changed AND height anomaly ≥ 0.5 m

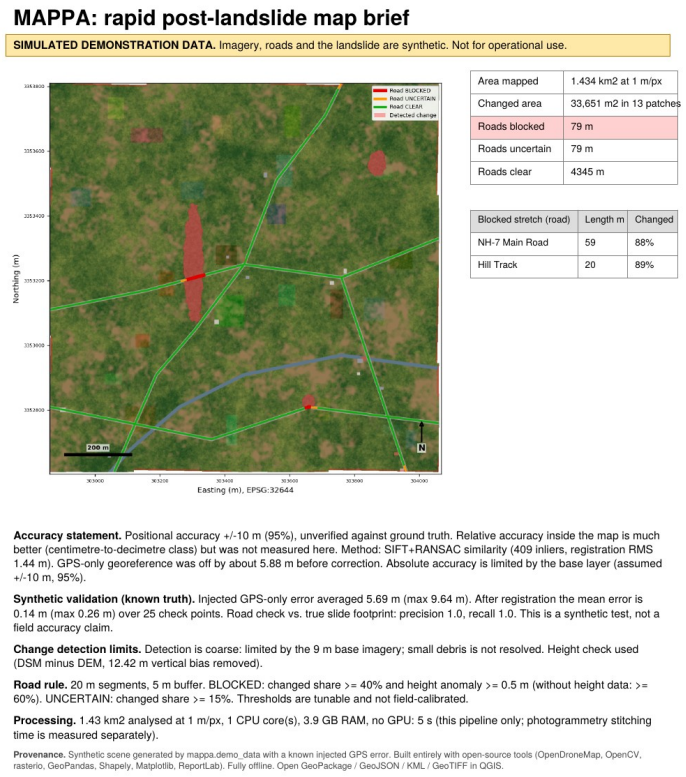
UNCERTAIN ≥ 15% changed, no height confirmation

CLEAR Little or no change

Change = colour difference after radiometric normalisation, plus DSM−DEM height with the vertical bias removed. Roads: 20 m segments, 5 m buffer. Thresholds are tunable and not field-calibrated.

SIMULATED DEMO SCENE

Result: a one-page brief responders can act on



MAPPA: rapid post-landslide map brief

SIMULATED DEMONSTRATION DATA. Imagery, roads and the landslide are synthetic. Not for operational use.

Area mapped	1.434 km2 at 1 m/px
Changed area	33,651 m2 in 13 patches
Roads blocked	79 m
Roads uncertain	79 m
Roads clear	4345 m

Blocked stretch (road)	Length m	Changed
NH-7 Main Road	59	88%
Hill Track	20	89%

Accuracy statement. Positional accuracy +/-10 m (95%), unverified against ground truth. Relative accuracy inside the map is much better (centimetre-to-decimetre class) but was not measured here. Method: SIFT+RANSAC similarity (409 inliers, registration RMS 1.44 m). GPS-only georeference was off by about 5.88 m before correction. Absolute accuracy is limited by the base layer (assumed +/-10 m, 95%).

Synthetic validation (known truth). Injected GPS-only error averaged 5.69 m (max 9.64 m). After registration the mean error is 0.14 m (max 0.26 m) over 25 check points. Road check vs. true slide footprint: precision 1.0, recall 1.0. This is a synthetic test, not a field accuracy claim.

Change detection limits. Detection is coarse: limited by the 9 m base imagery; small debris is not resolved. Height check used (DSM minus DEM, 12.42 m vertical bias removed).

Road rule. 20 m segments, 5 m buffer. BLOCKED: changed share >= 40% and height anomaly >= 0.5 m (without height data: >= 60%). UNCERTAIN: changed share >= 15%. Thresholds are tunable and not field-calibrated.

Processing. 1.43 km2 analysed at 1 m/px, 1 CPU core(s), 3.9 GB RAM, no GPU: 5 s (this pipeline only; photogrammetry stitching time is measured separately).

Provenance. Synthetic scene generated by mappa.demo_data with a known injected GPS error. Built entirely with open-source tools (OpenDroneMap, OpenCV, rasterio, GeoPandas, Shapely, Matplotlib, ReportLab). Fully offline. Open GeoPackage / GeoJSON / KML / GeoTIFF in QGIS.

79 m

Blocked road found: NH-7 Main Road 59 m, Hill Track 20 m

13

Change patches, 33,651 m², on a 1.43 km² scene

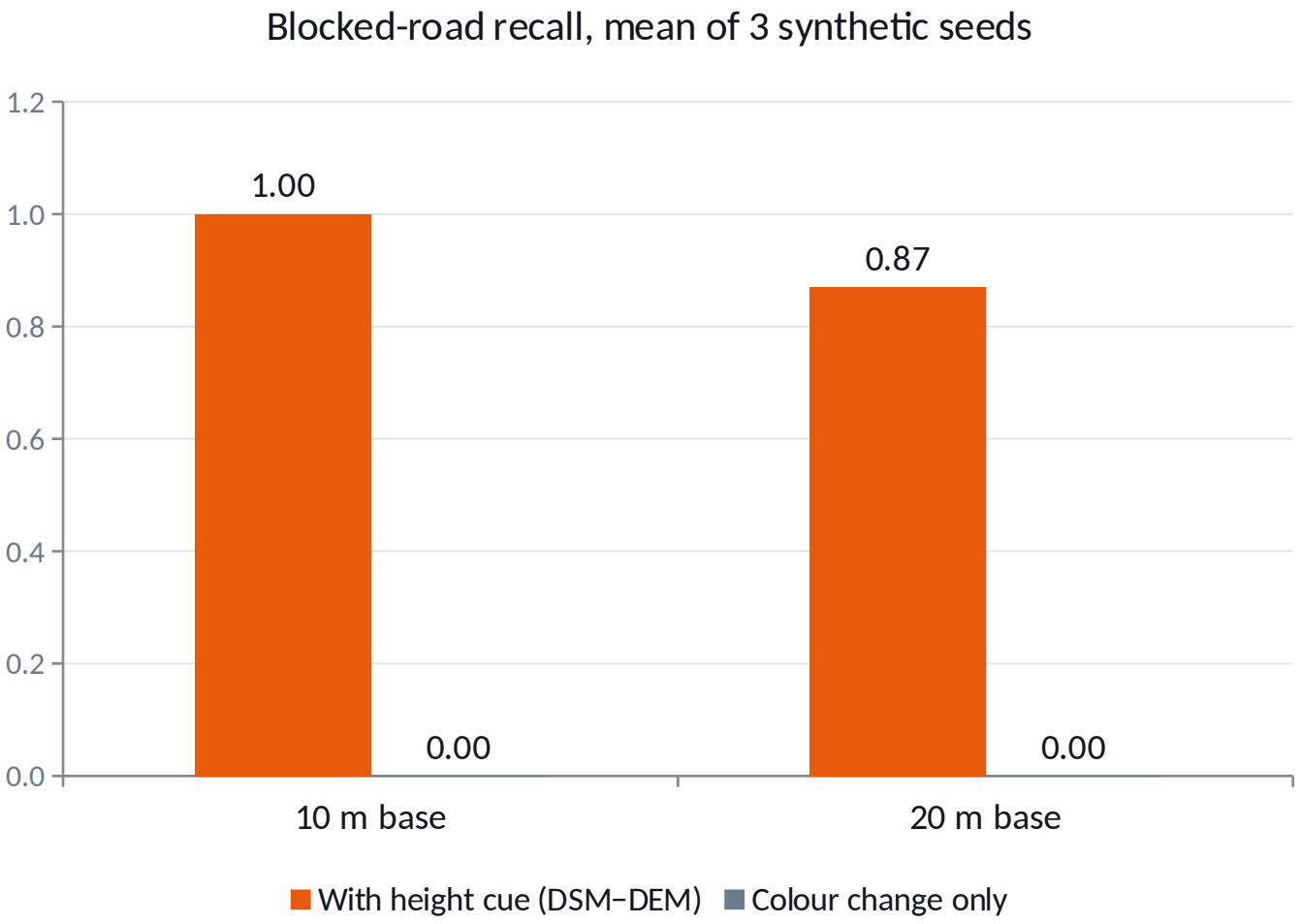
1.0 / 1.0

Road precision / recall vs known truth. Synthetic scene only.

Formats responders already open

GeoTIFF · GeoPackage · GeoJSON · KML · CSV · PDF. All 10 outputs re-opened by independent readers and verified.

Colour alone missed every blocked road



Why

Debris on a dirt road looks like the road. Height is the signal.

Impact

Precision stayed 1.0 with the cue. Colour-only gave false BLOCKED calls at a 20 m base.

Action

Always pre-cache a DEM and run ODM with --dsm.

Measured time, offline by design, all open source

5 s

1.43 km² at 1 m/px: register, change, roads, export, PDF. 1 CPU core, 3.9 GB RAM, no GPU.

Stitching time: to be measured on the target laptop

The 5 s excludes OpenDroneMap. scripts/run_odm.sh prints images, seconds and cores to quote. ODM was not run in the build environment (no Docker).

Open-source stack

- **OpenDroneMap**
  stitching, orthophoto, DSM
- **OpenCV**
  SIFT, RANSAC registration
- **rasterio / GDAL**
  raster IO, reprojection
- **GeoPandas, Shapely**
  road segments, vectors
- **OpenStreetMap, Sentinel-2, OAM, Copernicus DEM**
  base data
- **Matplotlib, ReportLab, QGIS**
  brief, map, viewing

Honest limits, and what we do next

Limits

- Validated on a simulated scene only, not yet on a real landslide
- Absolute accuracy is bounded by the base layer (about 10 m)
- Change detection is coarse at 10 m base resolution
- Thresholds are not field-calibrated; DSM−DEM also sees trees and buildings

Next steps

- Run ODM on OpenDroneMap sample sets and our own flight, and publish the measured time
- Test on real OpenAerialMap post-disaster imagery
- Calibrate thresholds against surveyed ground truth
- Add MBTiles and a QGIS project

github.com/kavixsri/MAPPA · MIT licensed · runs offline